

Finite-Time Feedback Control with Kinetic Compliance

Control, information, and irreversibility with a single measurable affinity

22 August 2026

Abstract

We develop a finite-state reference theory for feedback control of a binary population using one measurable directional affinity h and one kinetic compliance parameter γ . The derivation is deliberately finite-time and does not assume equilibrium or stationarity. Starting from the population ensemble $p_k(n)$, a hypergeometric sensor and stochastic feedback policy determine the advocacy probability. A reversible finite-compliance controlled kernel then yields an exact susceptibility, an exact state-local action-to-population information $T_\pi(n)$, and a closed mean controlled current. Its forward/reverse transition ratio produces a pathwise identity in which the finite-time system entropy change, the affinity-weighted current, and the sensing information sum exactly to the Kullback-Leibler irreversibility of the feedback path measure. Rather than introducing another bounded efficiency ratio, we characterize control by its current or susceptibility together with the finite-time irreversibility required to obtain it. The report also preserves an operational calibration from microscopic controlled LLM transitions and regenerates the state-local information sweeps under the single-affinity theory.

The conceptual chain used throughout is

$$\boxed{\text{sensing/policy} \longrightarrow \chi_{h,\gamma} \longrightarrow T_\pi(n) \longrightarrow J_c \longrightarrow \Sigma.} \quad (1)$$

Thermodynamic logarithms are natural logarithms, so entropy and irreversibility in the second-law identity are measured in nats. The state-local information $T_\pi(n)$ is reported in bits, matching the information-theoretic analysis convention used for the empirical action channel.

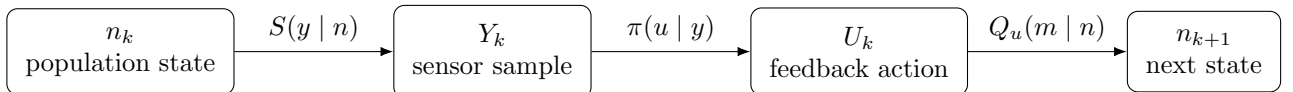
1 Problem and one-cycle architecture

Consider a population of N binary agents and fix a controller target Z . At feedback cycle k , let

$$\boxed{n_k \in \{0, 1, \dots, N\}, \quad x_k = \frac{n_k}{N},} \quad (2)$$

where n_k is the number of agents supporting Z immediately before sensing and control.

A single feedback cycle has the causal order



Equivalently,

$$\boxed{n_k \longrightarrow Y_k \xrightarrow{\pi} U_k \longrightarrow n_{k+1}.} \quad (3)$$

The controller policy is the conditional action law

$$\boxed{\pi(u \mid y) = P(U_k = u \mid Y_k = y)}. \quad (4)$$

The variables and parameters used below are:

N	population size
$n_k, x_k = n_k/N$	target count and target fraction before cycle k
q_c	sensor sample size: number of agents read by the controller
Y_k	number of target supporters in the sensor sample
$U_k \in \{0, 1\}$	feedback action: 0 = NoOp, 1 = advocate the target
θ	feedback threshold: location where the policy tends to switch
β	policy gain: sharpness of that switch
b	number of controlled microscopic opportunities when the controller acts
h	directional affinity: asymmetry favoring target-directed versus reverse controlled revisions
$\gamma \in [0, 1]$	kinetic compliance: how readily a controlled opportunity produces any revision
$\chi_{h,\gamma}(x)$	controller-induced susceptibility of the mean target fraction
$T_\pi(n)$	state-local action-to-population information
$J_{c,k}$	mean controlled target-count current in cycle k
Σ_k	finite-time path irreversibility for cycle k

Scope. The theory is a coarse-grained reference model for the controlled population coordinate n_k . It does not assign an energetic thermodynamics to the complete language-model population. The role of h is operational: it is the log directional odds of the controlled revision channel after the combinatorial population multiplicity is separated out. The role of γ is kinetic. This distinction is the central simplification of the model.

2 Population ensembles and finite-time notation

The population state is random across repeated realizations. If a derivation or theoretical figure begins from an initial ensemble, we write

$$\boxed{p_0(n) = P(n_0 = n)} \quad (5)$$

and call p_0 the *initial population-state distribution*.

At a generic cycle k ,

$$\boxed{p_k(n) = P(n_k = n)} \quad (6)$$

is the ensemble distribution before sensing and control, while

$$\boxed{p_{k+1}(m) = P(n_{k+1} = m)} \quad (7)$$

is the ensemble distribution after that feedback cycle. These distributions are not extra model parameters. They summarize the frequencies of population states across repeated realizations at the corresponding cycle.

For readability, a local one-cycle derivation may suppress time indices. Whenever that is done below,

$$\boxed{p(n) \equiv p_k(n), \quad p'(m) \equiv p_{k+1}(m)}. \quad (8)$$

No equilibrium or stationarity assumption is made. In general,

$$\boxed{p_{k+1} \neq p_k.} \quad (9)$$

The resulting finite-time change of the population ensemble is retained explicitly in the system entropy change derived later.

For the illustrative current-versus-irreversibility plot we will choose

$$\boxed{p_0(n) = \text{Binomial}(N, 0.25),} \quad (10)$$

only to provide a concrete theoretical ensemble. None of the identities below requires this particular choice.

3 Sensing and feedback decision

The controller samples q_c agents without replacement. If Y_k is the number of target supporters in the sample, then conditioned on $n_k = n$,

$$Y_k \mid n_k = n \sim \text{Hypergeom}(N, n, q_c), \quad (11)$$

with sensor kernel

$$S(y \mid n) = \frac{\binom{n}{y} \binom{N-n}{q_c-y}}{\binom{N}{q_c}}. \quad (12)$$

After observing $Y_k = y$, the controller advocates with probability

$$\boxed{\pi(1 \mid y) = \sigma \left[\beta \left(\theta - \frac{y}{q_c} \right) \right], \quad \sigma(z) = \frac{1}{1 + e^{-z}}.} \quad (13)$$

Thus θ sets the sensed target-share threshold around which the action changes, while β controls how sharply it changes.

The advocacy probability conditional on the true population state is obtained by averaging the policy over the sensor noise:

$$\boxed{a_n \equiv P(U_k = 1 \mid n_k = n) = \sum_y S(y \mid n) \pi(1 \mid y).} \quad (14)$$

This finite sum is where the sensing resource q_c and policy parameters (θ, β) enter the state-dependent action probability.

4 Finite-bias, finite-compliance microscopic actuation

Use one directional affinity h and define

$$p_h = \sigma(h). \quad (15)$$

At one controlled microscopic opportunity, a focal agent is sampled uniformly. If the focal agent is currently non-target, it switches to the target with probability γp_h . If it is currently target, it switches away with probability $\gamma(1 - p_h)$. Therefore

$$K_{h,\gamma}(n+1 \mid n) = \gamma \frac{N-n}{N} p_h, \quad (16)$$

$$K_{h,\gamma}(n-1 \mid n) = \gamma \frac{n}{N} (1 - p_h), \quad (17)$$

$$K_{h,\gamma}(n \mid n) = 1 - K_{h,\gamma}(n+1 \mid n) - K_{h,\gamma}(n-1 \mid n). \quad (18)$$

The two parameters have deliberately different jobs:

$$\boxed{h = \text{directional asymmetry}, \quad \gamma = \text{kinetic compliance.}} \quad (19)$$

The affinity h controls which direction is preferred when a controlled revision occurs. The compliance γ controls how readily a controlled opportunity produces a revision at all.

5 Exact susceptibility: derivation of $\chi_{h,\gamma}(x)$

We now derive the exact mean response produced by b controlled opportunities. The result is

$$\boxed{\chi_{h,\gamma}(x) = [\sigma(h) - x]\Lambda_{b,\gamma}, \quad \Lambda_{b,\gamma} = 1 - \left(1 - \frac{\gamma}{N}\right)^b.} \quad (20)$$

5.1 One controlled opportunity

Suppose the target fraction immediately before the j th controlled opportunity is x_j . The count changes by $+1$ with probability

$$P(\Delta n = +1 \mid x_j) = \gamma(1 - x_j)p_h, \quad (21)$$

and by -1 with probability

$$P(\Delta n = -1 \mid x_j) = \gamma x_j(1 - p_h). \quad (22)$$

Consequently,

$$\mathbb{E}[\Delta n \mid x_j] = \gamma(1 - x_j)p_h - \gamma x_j(1 - p_h) \quad (23)$$

$$= \gamma[p_h - x_j]. \quad (24)$$

This is already useful: the expected microscopic current points toward $p_h = \sigma(h)$.

Since $x = n/N$,

$$\mathbb{E}[\Delta x \mid x_j] = \frac{\gamma}{N}[p_h - x_j]. \quad (25)$$

Hence

$$\mathbb{E}[x_{j+1} \mid x_j] = x_j + \frac{\gamma}{N}(p_h - x_j) \quad (26)$$

$$= \left(1 - \frac{\gamma}{N}\right) x_j + \frac{\gamma}{N} p_h. \quad (27)$$

Define

$$\alpha_\gamma = 1 - \frac{\gamma}{N}. \quad (28)$$

Then

$$\mathbb{E}[x_{j+1} \mid x_j] = \alpha_\gamma x_j + (1 - \alpha_\gamma)p_h. \quad (29)$$

5.2 Iterating exactly b controlled opportunities

Let $m_j = \mathbb{E}[x_j \mid x_0 = x]$. Taking expectations of the affine recursion gives

$$m_{j+1} = \alpha_\gamma m_j + (1 - \alpha_\gamma)p_h. \quad (30)$$

Subtract the fixed point p_h :

$$m_{j+1} - p_h = \alpha_\gamma(m_j - p_h). \quad (31)$$

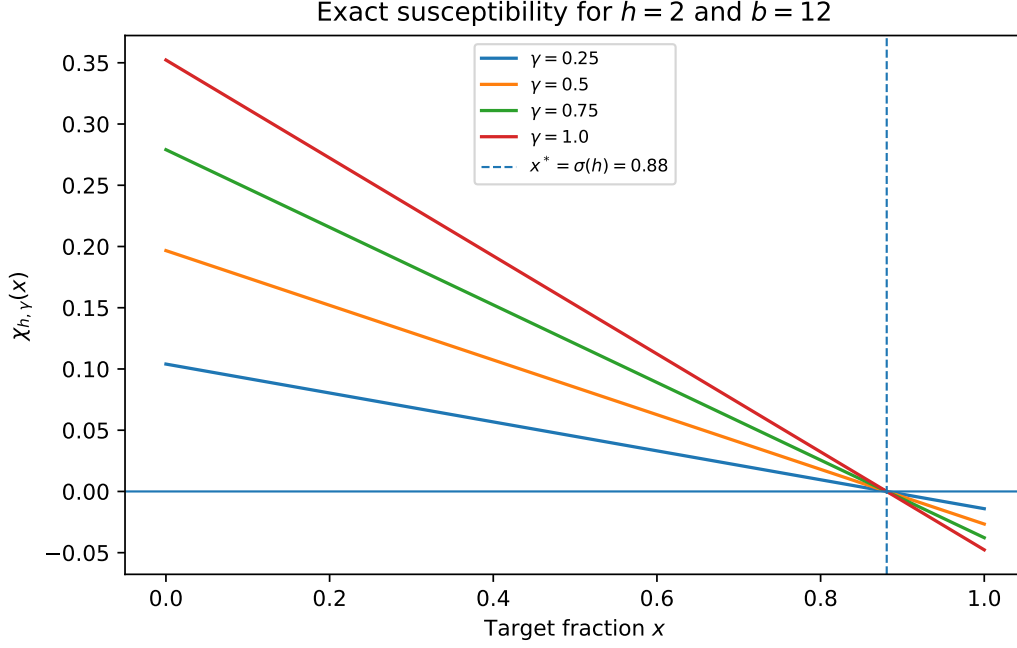


Figure 1: Exact susceptibility for $h = 2$ and $b = 12$. Finite kinetic compliance changes the response amplitude through $\Lambda_{b,\gamma}$ without moving the zero-response point $x^* = \sigma(h)$.

Iterating b times,

$$m_b - p_h = \alpha_\gamma^b (x - p_h), \quad (32)$$

so

$$m_b = p_h + (x - p_h) \alpha_\gamma^b \quad (33)$$

$$= x + (p_h - x)(1 - \alpha_\gamma^b). \quad (34)$$

Therefore

$$\mathbb{E}[x_b - x \mid x] = (p_h - x) \left[1 - \left(1 - \frac{\gamma}{N} \right)^b \right]. \quad (35)$$

If ordinary $q = 1$ voter updates are interspersed between controlled opportunities, their conditional mean obeys

$$\mathbb{E}[x_{j+1} \mid x_j, \text{ordinary}] = x_j. \quad (36)$$

They are therefore mean-neutral and do not change this exact advocacy-versus-NoOp mean response. We define

$$\chi_{h,\gamma}(x) \equiv \mathbb{E}[x' \mid U = 1, x] - \mathbb{E}[x' \mid U = 0, x] = [\sigma(h) - x] \Lambda_{b,\gamma}. \quad (37)$$

The directional affinity fixes the preferred set point, while kinetic compliance controls the response amplitude. The zero-response point is

$$x^* = \sigma(h). \quad (38)$$

6 Exact state-local action-to-population information

The susceptibility measures a signed mean response. A complementary question is whether the controller action is statistically visible in the next population state. This is captured by the exact state-local action-to-population information.

6.1 Definition

For a fixed pre-control state $n = n_k$, use the shorter one-cycle notation

$$m \equiv n_{k+1}. \quad (39)$$

Define the two controlled-layer next-state kernels

$$Q_0(m | n) = \delta_{mn}, \quad Q_1(m | n) = K_{h,\gamma}^b(m | n). \quad (40)$$

With $a_n = P(U_k = 1 | n_k = n)$, the action-marginalized kernel is

$$Q_\pi(m | n) = (1 - a_n)Q_0(m | n) + a_nQ_1(m | n). \quad (41)$$

The state-local action-to-population information is

$$T_\pi(n) = I(U_k; n_{k+1} | n_k = n) = \sum_{u \in \{0,1\}} P(u | n) \sum_m Q_u(m | n) \log_2 \frac{Q_u(m | n)}{Q_\pi(m | n)}. \quad (42)$$

Thus, in the local notation $m = n_{k+1}$, this is exactly $I(U; m | n)$.

The subscript π emphasizes that the information channel is induced by the feedback policy. The quantity is large only when two conditions hold simultaneously:

1. the action remains variable at that state, so a_n is not too close to 0 or 1;
2. the ADVOCATE and NoOp next-state kernels are sufficiently separated.

If either condition fails, observing the next population state reveals little about whether the controller acted.

If an ensemble-weighted quantity is desired at cycle k , then

$$I(U_k; n_{k+1} | n_k) = \sum_n p_k(n) T_\pi(n). \quad (43)$$

The summary figure below instead uses the simple arithmetic state average $(N + 1)^{-1} \sum_n T_\pi(n)$ so that it describes the channel geometry without choosing an occupancy distribution.

Do not confuse the two information quantities. The sensing information introduced later in the finite-time second law is

$$I(n_k; Y_k) = \text{sensing information}, \quad (44)$$

whereas

$$T_\pi(n) = I(U_k; n_{k+1} | n_k = n) = \text{action-to-population information}. \quad (45)$$

The first quantifies what the sensor learns about the current population; the second quantifies how visible the action is in the next population state.

6.2 Baseline landscape

Figure 2 shows the exact local landscape as a function of the current target fraction $x = n/N$ and actuation budget b , for the baseline values $N = 24$, $h = 2$, $q_c = 12$, $\beta = 4$, $\theta = 1/2$, and $\gamma = 1$.

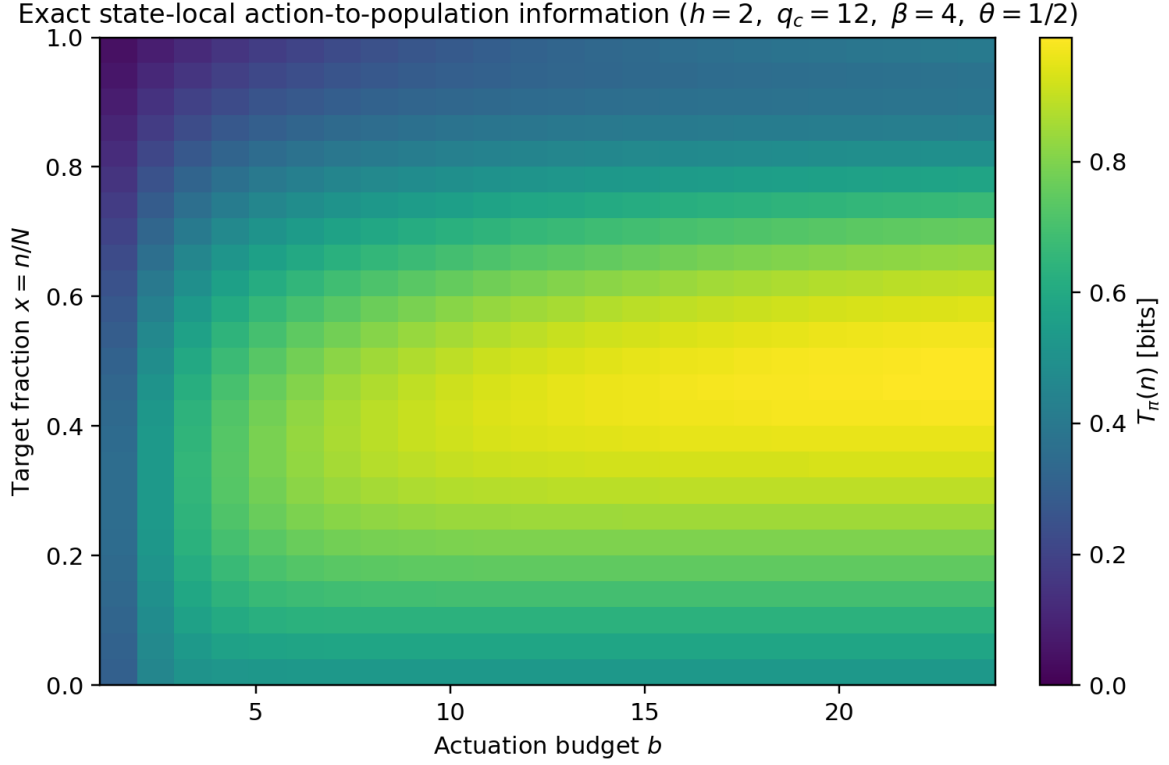


Figure 2: Exact state-local action-to-population information for $N = 24$, $h = 2$, $q_c = 12$, $\beta = 4$, $\theta = 1/2$, and $\gamma = 1$. Increasing b separates the advocacy and NoOp kernels more strongly, but information remains small in states where the policy action is nearly deterministic.

6.3 Parameter sweeps

Figures 3–6 vary h , q_c , β , and θ one at a time while holding the other parameters at their baseline values.

6.4 How to read the parameter dependence

The same interpretation as in the state-local picture is useful throughout. Increasing h primarily changes *kernel separation*. Changing q_c , β , or θ primarily changes *action variability through sensing and policy*. Large $T_\pi(n)$ requires both. This is why a stronger controller does not produce a uniformly larger information channel at every state: if a policy becomes deterministic, its action entropy can collapse even when its action has a strong behavioral effect.

7 Controlled current

We now convert the susceptibility into a target-directed controlled current. Define

$$Q_1 = K_{h,\gamma}^b, \quad Q_0 = I. \quad (46)$$

For one cycle, the controlled current increment is the signed target-count displacement when the controller acts,

$$j_c = U_k(n_{k+1} - n_k). \quad (47)$$

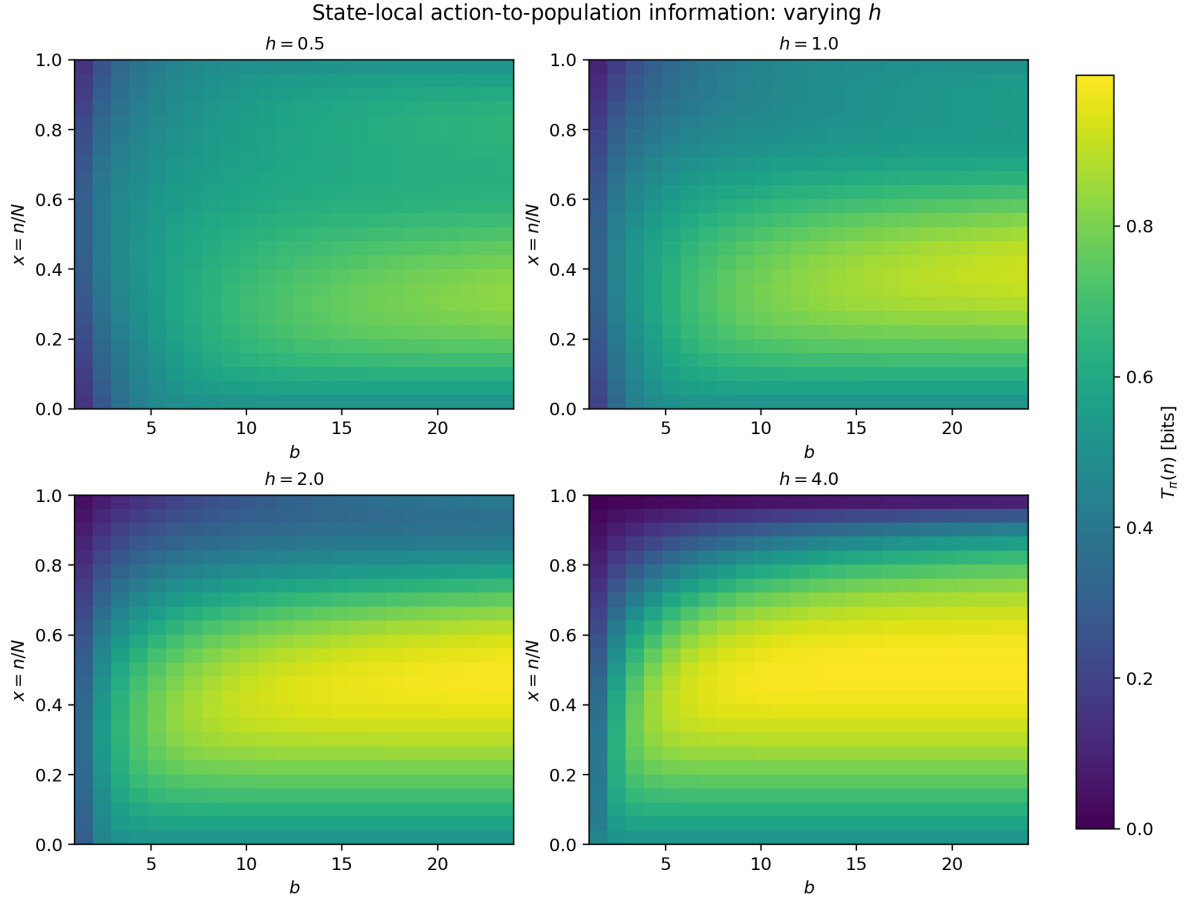


Figure 3: Exact state-local action-to-population information while varying the directional affinity h . Larger h changes and generally strengthens separation of the advocacy kernel from NoOp, with the enhancement concentrated where the policy retains nontrivial action entropy.

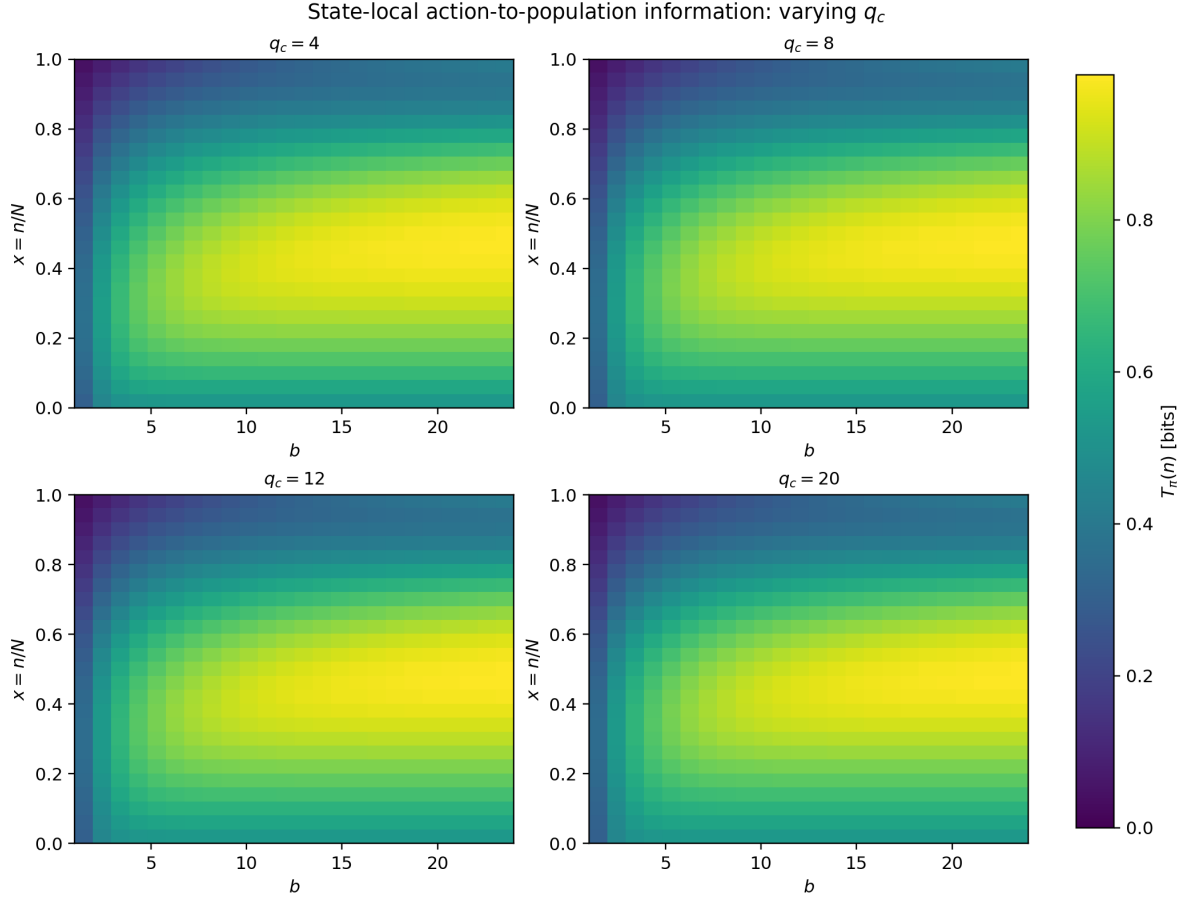


Figure 4: Exact state-local action-to-population information while varying q_c . The sensor sample size changes how sharply the controller resolves the current target fraction, thereby changing the state dependence of a_n and the ridge structure of $T_\pi(n)$.

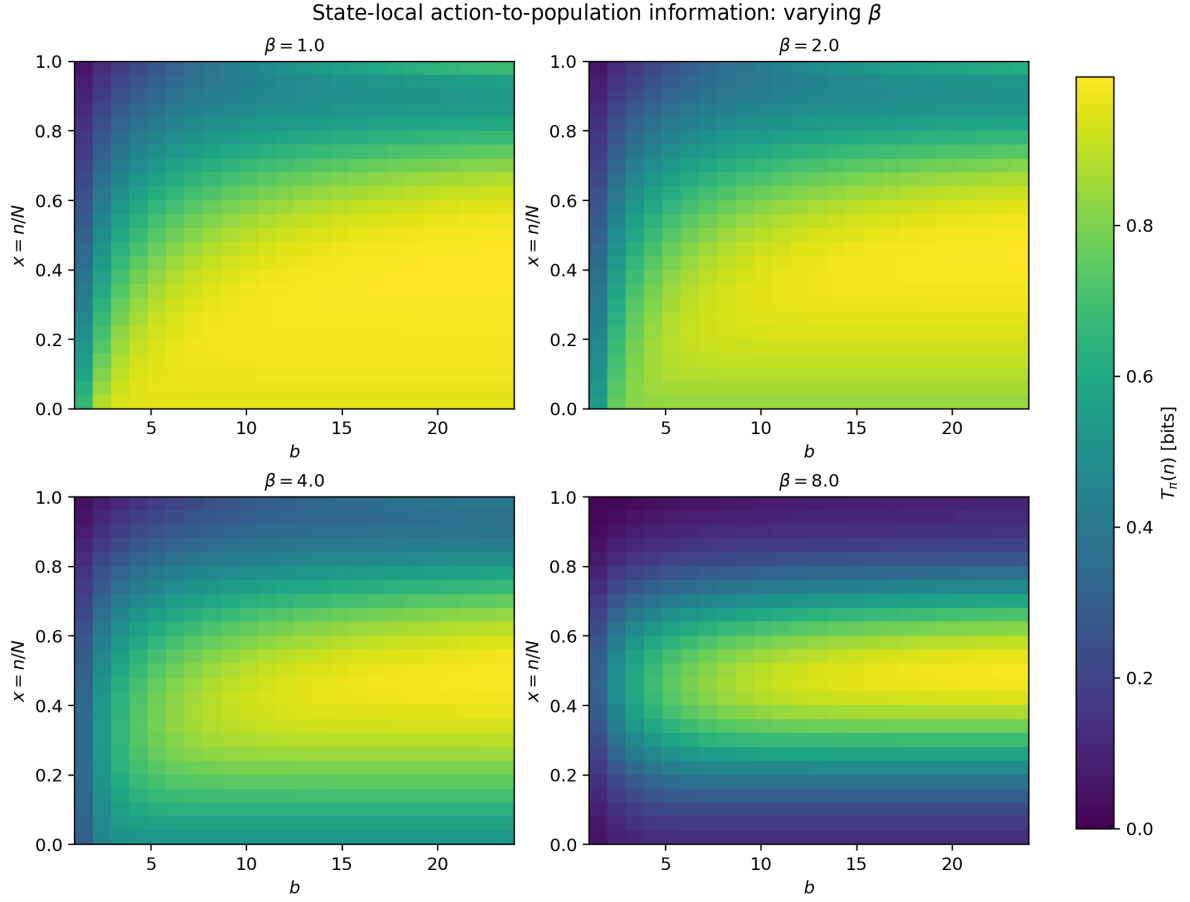


Figure 5: Exact state-local action-to-population information while varying β . Larger β makes the policy more switch-like: this can sharpen the informative band around the switching region while suppressing information where the action saturates.

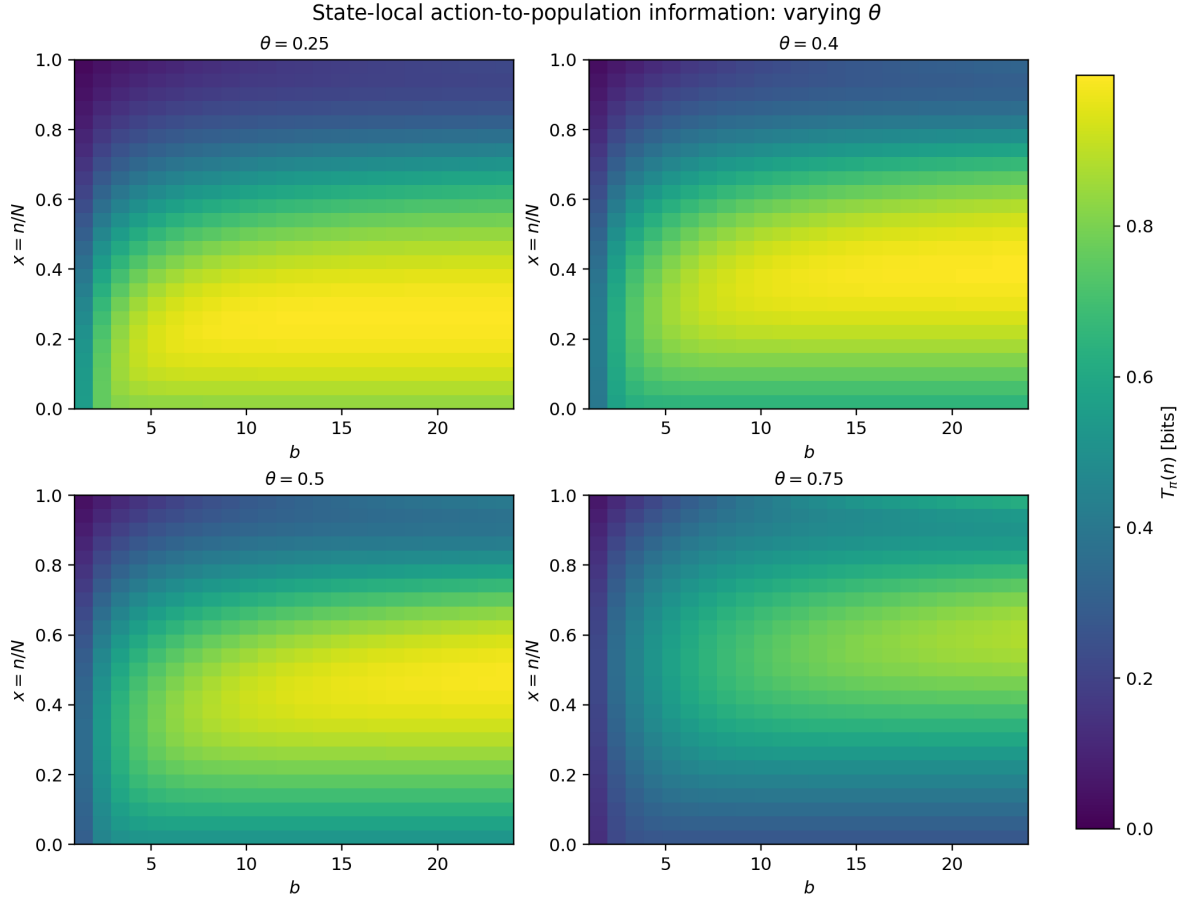


Figure 6: Exact state-local action-to-population information while varying θ . The threshold shifts the state-space region where the controller is undecided and therefore shifts the high-information ridge.

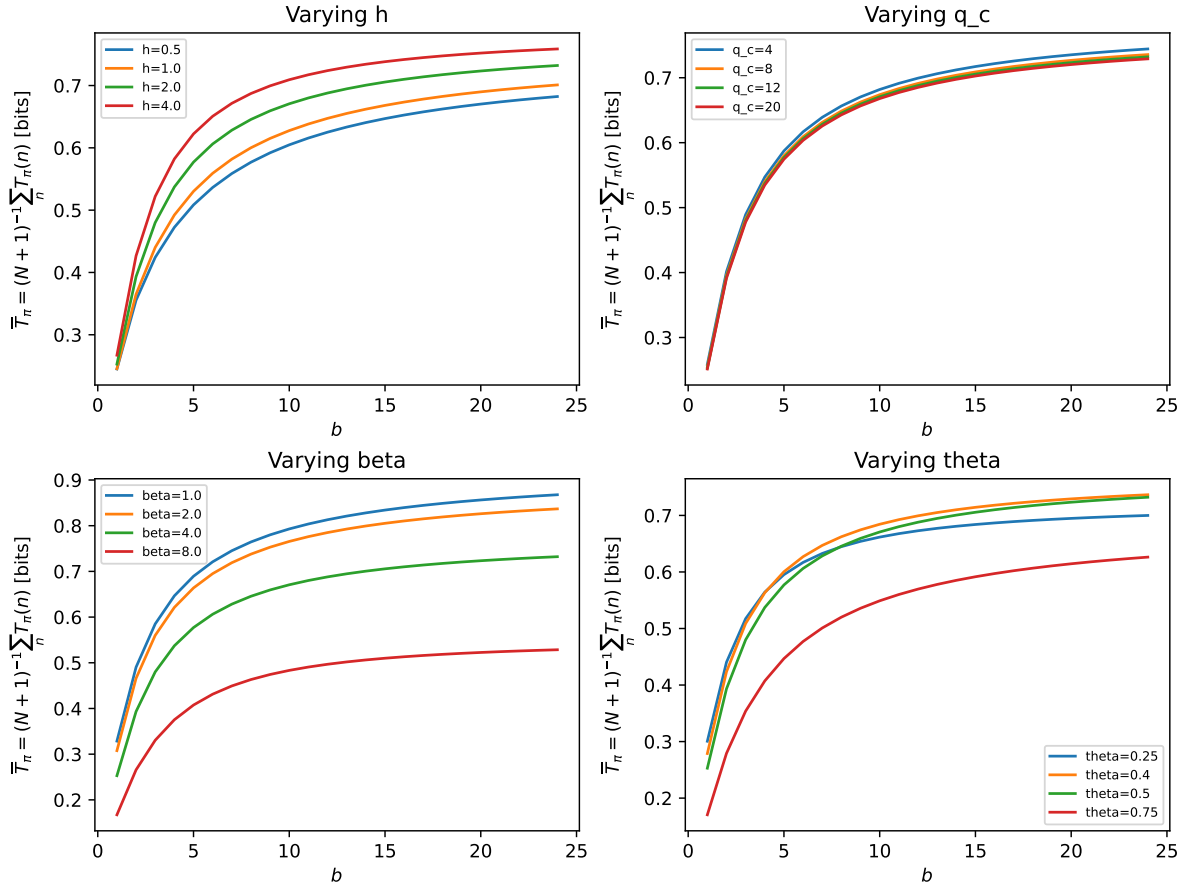


Figure 7: Arithmetic state-average $\bar{T}_\pi = (N+1)^{-1} \sum_n T_\pi(n)$ for the four parameter sweeps. This compact view complements the heatmaps; it does not assume a population ensemble.

Conditioned on state n and advocacy,

$$\mathbb{E}[j_c \mid n_k = n, U_k = 1] = N\chi_{h,\gamma}(n/N) \quad (48)$$

$$= N\Lambda_{b,\gamma} \left[\sigma(h) - \frac{n}{N} \right]. \quad (49)$$

The action is only taken with probability a_n , and the pre-sensing state is distributed as $p_k(n)$. Averaging therefore gives

$$J_{c,k} = N\Lambda_{b,\gamma} \sum_n p_k(n) a_n \left[\sigma(h) - \frac{n}{N} \right]. \quad (50)$$

When the cycle index is suppressed we write $J_c \equiv J_{c,k}$. Positive J_c means that controlled revisions produce a net movement toward the controller target on average; negative J_c means the net controlled movement is away from it.

8 Transition-ratio identity

The key algebraic property is that γ changes the kinetics but cancels from the directional forward/reverse ratio. For neighboring count states,

$$\frac{K_{h,\gamma}(n+1 \mid n)}{K_{h,\gamma}(n \mid n+1)} = \frac{\gamma \frac{N-n}{N} p_h}{\gamma \frac{n+1}{N} (1-p_h)} \quad (51)$$

$$= \frac{N-n}{n+1} \frac{p_h}{1-p_h} \quad (52)$$

$$= \frac{N-n}{n+1} e^h. \quad (53)$$

Define the combinatorial macrostate entropy

$$S_{\text{mix}}(n) = \ln \binom{N}{n}. \quad (54)$$

Because

$$S_{\text{mix}}(n+1) - S_{\text{mix}}(n) = \ln \frac{N-n}{n+1}, \quad (55)$$

we obtain

$$\ln \frac{K_{h,\gamma}(n+1 \mid n)}{K_{h,\gamma}(n \mid n+1)} = S_{\text{mix}}(n+1) - S_{\text{mix}}(n) + h. \quad (56)$$

The compliance factor cancels exactly. Directionality and kinetics therefore remain distinct even at the path level.

Define the positive weight

$$w_h(n) = \binom{N}{n} e^{hn}. \quad (57)$$

The neighboring-state relation is equivalent to

$$w_h(n) K_{h,\gamma}(m \mid n) = w_h(m) K_{h,\gamma}(n \mid m) \quad (58)$$

for every supported transition. For diagonal transitions the equality is immediate; for $m = n \pm 1$ it follows from the ratio just derived.

This weighted reversibility relation is preserved by matrix multiplication. Therefore, with $Q_1 = K_{h,\gamma}^b$,

$$w_h(n) Q_1(m \mid n) = w_h(m) Q_1(n \mid m), \quad (59)$$

and hence

$$\ln \frac{Q_1(m | n)}{Q_1(n | m)} = S_{\text{mix}}(m) - S_{\text{mix}}(n) + h(m - n). \quad (60)$$

This endpoint identity is what allows the finite-time path derivation to be written without resolving every controlled microscopic jump.

9 Forward and reverse finite-time feedback paths

For feedback cycle k , the forward path measure over the variables (n, y, u, m) is

$$P_F^{(k)}(n, y, u, m) = p_k(n) S(y | n) \pi(u | y) Q_u(m | n). \quad (61)$$

The forward final population distribution is obtained by marginalization:

$$p_{k+1}(m) = \sum_{n, y, u} P_F^{(k)}(n, y, u, m). \quad (62)$$

Equivalently, after averaging over the sensor and policy,

$$p_{k+1}(m) = \sum_n p_k(n) [(1 - a_n) \delta_{mn} + a_n Q_1(m | n)]. \quad (63)$$

The sensor marginal at the beginning of the cycle is

$$p_{Y,k}(y) = \sum_n p_k(n) S(y | n). \quad (64)$$

Use the reverse reference experiment

$$P_R^{(k)}(m, y, u, n) = p_{k+1}(m) p_{Y,k}(y) \pi(u | y) Q_u(n | m). \quad (65)$$

This construction has a direct operational reading. The reverse reference starts from the *forward final ensemble* p_{k+1} , samples a stored sensor label y independently from its forward marginal $p_{Y,k}$, applies the same policy law $\pi(u | y)$, and uses the reverse of the same controlled transition kernel. Because the same policy factor appears in the forward and reverse path weights, it cancels from their ratio.

Crucially, the reverse experiment does not assume stationarity. It starts from p_{k+1} whether or not that distribution equals p_k .

10 Finite-time path irreversibility and second-law identity

Define the stochastic path log-ratio

$$\Sigma_k^* = \ln \frac{P_F^{(k)}(n, y, u, m)}{P_R^{(k)}(m, y, u, n)}. \quad (66)$$

Substituting the path measures gives

$$\Sigma_k^* = \ln \frac{p_k(n)}{p_{k+1}(m)} + \ln \frac{S(y | n)}{p_{Y,k}(y)} + \ln \frac{Q_u(m | n)}{Q_u(n | m)}. \quad (67)$$

The factor $\pi(u | y)$ cancels exactly.

Define the pointwise sensing information

$$i_k(n; y) = \ln \frac{S(y | n)}{p_{Y,k}(y)}, \quad (68)$$

and the stochastic system entropy

$$s_{\text{sys}}(n; p_k) = -\ln p_k(n) + S_{\text{mix}}(n). \quad (69)$$

The controlled current increment is

$$j_c = u(m - n). \quad (70)$$

For $u = 0$, $Q_0 = I$, so $m = n$ and $j_c = 0$. For $u = 1$, the transition-ratio identity applies. Therefore

$$\ln \frac{p_k(n)}{p_{k+1}(m)} + S_{\text{mix}}(m) - S_{\text{mix}}(n) = s_{\text{sys}}(m; p_{k+1}) - s_{\text{sys}}(n; p_k), \quad (71)$$

and we obtain the central pathwise identity

$$\Sigma_k^* = \Delta s_{\text{sys}} + h j_c + i_k(n; y). \quad (72)$$

Average over the forward path ensemble. By construction,

$$\Sigma_k \equiv \mathbb{E}_F[\Sigma_k^*] = D_{\text{KL}}(P_F^{(k)} \| P_R^{(k)}) \geq 0. \quad (73)$$

The three average terms are

$$\Delta S_{\text{sys},k} = S_{\text{sys}}[p_{k+1}] - S_{\text{sys}}[p_k], \quad (74)$$

$$J_{c,k} = \mathbb{E}_F[j_c], \quad (75)$$

$$I(n_k; Y_k) = \mathbb{E}_F[i_k(n; Y)]. \quad (76)$$

The system-entropy functional is

$$S_{\text{sys}}[p] = -\sum_n p(n) \ln p(n) + \sum_n p(n) \ln \binom{N}{n}. \quad (77)$$

The first term is the Shannon entropy of the coarse population state and the second averages the internal combinatorial multiplicity of each macrostate.

The sensing information has the finite-sum form

$$I(n_k; Y_k) = \sum_{n,y} p_k(n) S(y | n) \ln \frac{S(y | n)}{p_{Y,k}(y)}. \quad (78)$$

Combining the averages yields the main finite-time stochastic-thermodynamic statement:

$$\Delta S_{\text{sys},k} + h J_{c,k} + I(n_k; Y_k) = \Sigma_k = D_{\text{KL}}(P_F^{(k)} \| P_R^{(k)}) \geq 0. \quad (79)$$

This is a statement about finite-time path irreversibility of the coarse-grained feedback process. It is not a work balance. No equilibrium state, steady state, heat bath, or energetic load is required for the identity.

The finite-time entropy change is essential. If the controller shifts the population ensemble during a cycle, then $p_{k+1} \neq p_k$ and $\Delta S_{\text{sys},k}$ need not vanish. Dropping this term would silently impose a condition that the derivation does not need.

11 Control versus irreversibility

The natural design question in the present model is not a work-efficiency ratio. It is instead:

How much finite-time irreversibility is required to obtain a given amount of population control?

A controller configuration can therefore be characterized by the pair

$$\boxed{(J_c, \Sigma)} \quad (80)$$

or, when the response at a particular state is the relevant observable, by

$$\boxed{(\chi, \Sigma)}. \quad (81)$$

A direct control-design formulation is

$$\boxed{\pi^* = \arg \min_{\pi} \Sigma(\pi) \quad \text{subject to} \quad J_c(\pi) \geq J_{\text{req}},} \quad (82)$$

or equivalently

$$\boxed{\pi^* = \arg \max_{\pi} J_c(\pi) \quad \text{subject to} \quad \Sigma(\pi) \leq \Sigma_{\text{max}}.} \quad (83)$$

These are Pareto formulations: they do not require manufacturing a new bounded efficiency number.

Figure 8 illustrates this for one finite-time ensemble. It uses

$$p_0(n) = \text{Binomial}(24, 0.25), \quad (84)$$

with $h = 2$, $\gamma = 0.75$, $\beta = 4$, and $\theta = 1/2$. Each curve varies the actuation budget $b = 1, \dots, 24$, while the curves compare different sensor sizes q_c . The corresponding data file also records the direct KL calculation and the decomposition $\Delta S_{\text{sys}} + hJ_c + I(n; Y)$; the two agree numerically to floating-point precision.

12 Direct finite-sum computation recipe

For one cycle with pre-sensing ensemble $p_k(n)$, every theoretical quantity above is computable by finite sums and a finite matrix power:

1. Compute the sensor kernel $S(y | n)$ and policy $\pi(1 | y)$.
2. Compute

$$a_n = \sum_y S(y | n) \pi(1 | y), \quad p_{Y,k}(y) = \sum_n p_k(n) S(y | n).$$

3. Set $p_h = \sigma(h)$ and

$$\Lambda_{b,\gamma} = 1 - \left(1 - \frac{\gamma}{N}\right)^b.$$

4. Build $K_{h,\gamma}$ and $Q_1 = K_{h,\gamma}^b$, with $Q_0 = I$.

5. Compute the mean controlled current

$$J_{c,k} = N \Lambda_{b,\gamma} \sum_n p_k(n) a_n \left[p_h - \frac{n}{N} \right].$$

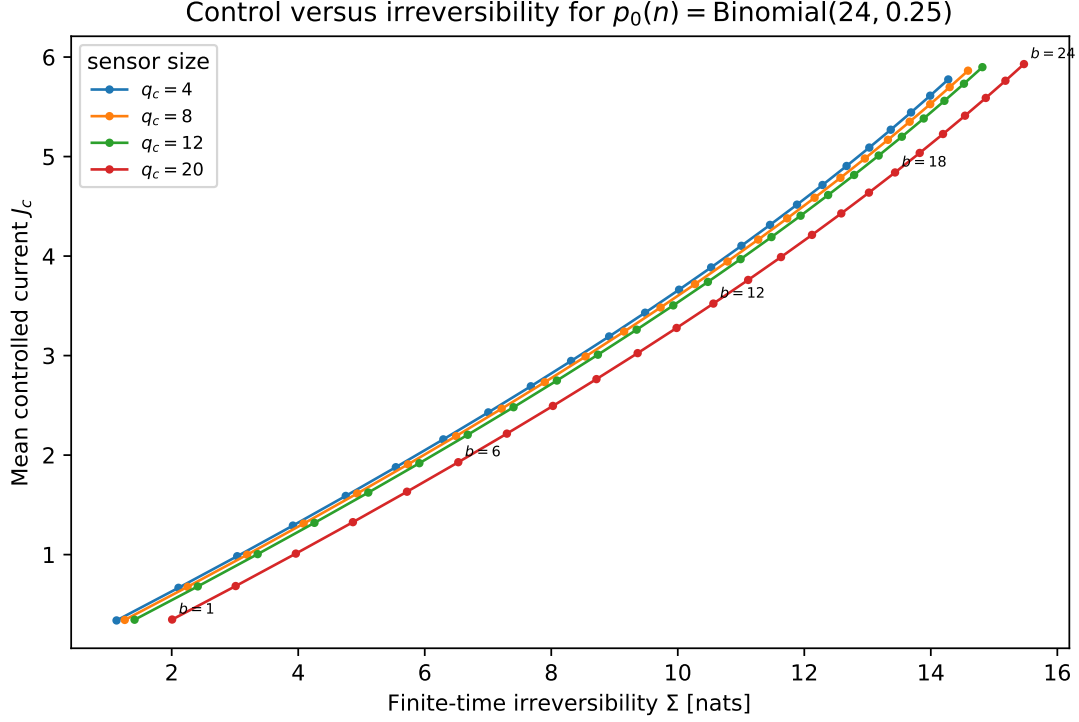


Figure 8: Mean controlled current versus finite-time irreversibility for an illustrative initial ensemble $p_0(n) = \text{Binomial}(24, 0.25)$. Each curve varies b from 1 to 24 at fixed $h = 2$, $\gamma = 0.75$, $\beta = 4$, and $\theta = 1/2$; labels mark selected b values. Different sensing resources can yield nearly the same controlled current while requiring different total path irreversibility because the sensing-information contribution changes.

6. Compute the sensing information

$$I(n_k; Y_k) = \sum_{n,y} p_k(n) S(y | n) \ln \frac{S(y | n)}{p_{Y,k}(y)}.$$

7. Propagate the ensemble exactly:

$$p_{k+1}(m) = \sum_n p_k(n) [(1 - a_n) \delta_{mn} + a_n Q_1(m | n)].$$

8. Compute

$$\Delta S_{\text{sys},k} = S_{\text{sys}}[p_{k+1}] - S_{\text{sys}}[p_k].$$

9. Finally evaluate

$$\boxed{\Sigma_k = \Delta S_{\text{sys},k} + h J_{c,k} + I(n_k; Y_k).}$$

As a numerical check, compute the path measures directly and verify

$$\Sigma_k = D_{\text{KL}}(P_F^{(k)} \| P_R^{(k)}).$$

There is no Monte Carlo approximation in this reference calculation. Monte Carlo or direct counting enters only when the same objects are estimated from LLM trajectories rather than generated by the finite-state reference kernel.

12.1 How to estimate the terms from repeated trajectories

At a fixed cycle k , $p_k(n)$ and $p_{k+1}(m)$ can be estimated as empirical histograms across repeated episodes. The sensor kernel is known exactly under the sampling protocol, while the logged policy probabilities or action frequencies estimate a_n . The controlled current can be estimated directly as the sample mean of $U_k(n_{k+1} - n_k)$, or from the calibrated reference kernel. Sensing information is a contingency-table mutual information between n_k and Y_k ; if an empirical analysis reports it in bits, multiply by $\ln 2$ before inserting it into the natural-log thermodynamic identity. The state-local $T_\pi(n)$ is a separate conditional mutual information and can remain in bits for comparison with the empirical action channel.

Estimating the full path KL directly from sparse LLM data is more demanding because it requires support for both forward and reverse coarse-grained path probabilities. The equality above therefore provides a useful model-based decomposition, while empirical support and overlap diagnostics remain necessary when conditional channels are estimated by direct counting.

13 Role of kinetic compliance

The kinetic parameter γ controls the speed and amplitude of the response, not its directional affinity. At the mean-response level it appears through

$$\Lambda_{b,\gamma} = 1 - \left(1 - \frac{\gamma}{N}\right)^b. \quad (85)$$

At the path level it changes $Q_1 = K_{h,\gamma}^b$ and therefore changes the final ensemble p_{k+1} and the path KL divergence. But it cancels from the forward/reverse microscopic ratio because the same factor multiplies both directions. There is therefore no additional $\ln \gamma$ contribution to the directional affinity.

At $\gamma = 0$, $K_{h,0} = I$ and $J_c = 0$: the controller has no kinetic actuation. For $0 < \gamma \leq 1$, the full finite-time derivation applies. In the large- h limit, directionality becomes nearly one-sided, but actuation can still remain incomplete when $\gamma < 1$.

14 Operational calibration from microscopic LLM transitions

Controlled microscopic transitions provide a direct operational calibration of both model parameters. Define

$$p_+ = P(\text{non-}Z \rightarrow Z \mid \text{controlled}), \quad (86)$$

$$p_- = P(Z \rightarrow \text{non-}Z \mid \text{controlled}). \quad (87)$$

Under the finite-compliance kernel,

$$p_+ = \gamma \sigma(h), \quad p_- = \gamma \sigma(-h). \quad (88)$$

Adding the two probabilities gives

$$p_+ + p_- = \gamma [\sigma(h) + \sigma(-h)] = \gamma, \quad (89)$$

while their ratio gives

$$\frac{p_+}{p_-} = \frac{\sigma(h)}{\sigma(-h)} = e^h. \quad (90)$$

Hence

$$\boxed{\gamma_{\text{eff}} = p_+ + p_-, \quad h_{\text{eff}} = \ln \frac{p_+}{p_-}.} \quad (91)$$

Using the existing controlled-transition counts,

$$p_+ = \frac{208}{572} \approx 0.364, \quad p_- = \frac{4}{508} \approx 0.0079, \quad (92)$$

we recover

$$\boxed{\gamma_{\text{eff}} \approx 0.372, \quad h_{\text{eff}} \approx 3.83.} \quad (93)$$

This is a complete operational calibration of the two parameters in the simplified controlled kernel: one number describes kinetic compliance and the other describes directional asymmetry.

15 Scope and modeling limitations

The theory should be read as a finite-state coarse-grained reference model for the controlled population coordinate n_k . It is not a claim of a complete thermodynamics of all social interactions among language-model agents.

The derivation is explicitly finite-time. It does not assume equilibrium or stationarity. The changing population ensemble is represented by

$$p_k \longrightarrow p_{k+1}, \quad (94)$$

and contributes

$$\Delta S_{\text{sys},k} = S_{\text{sys}}[p_{k+1}] - S_{\text{sys}}[p_k]. \quad (95)$$

This is why the reverse reference begins from the forward final ensemble rather than from the initial ensemble.

Finally, n_k is a coarse-grained state variable. In the LLM system, future transitions may depend on hidden epistemic state, private fact sets, dialogue history, persistent reasons, or other variables not encoded in the target count. The empirical reduced process in n_k can therefore be imperfectly Markovian. This is a modeling limitation to diagnose, not an assumption to hide. The finite-state kernel should consequently be used as a matched reference model whose deviations from the LLM trajectories are scientifically informative.

What each observable means. The sensing term $I(n_k; Y_k)$ asks how much the controller learns about the current population. The susceptibility $\chi_{h,\gamma}(x)$ asks for the signed mean response at a given state. The local information $T_\pi(n)$ asks how much the next population state reveals about the controller action. The current $J_{c,k}$ aggregates the target-directed response over the occupied population states. Finally, Σ_k quantifies the finite-time statistical irreversibility of the complete coarse-grained feedback cycle relative to the specified reverse reference. Together these quantities separate sensing, action visibility, macroscopic control, and irreversibility rather than compressing them into one scalar efficiency.

References

- [1] J. M. Horowitz and H. Sandberg, “Second-law-like inequalities with information and their interpretations,” *New Journal of Physics*, 16, 125007 (2014).
- [2] L. Irisarri, L. Trigel, R. Toral, and G. Manzano, “Stochastic Thermodynamics of Social Imitation beyond Energetics,” 2026.